Transport Protocols & User Datagram Protocol (UDP)

Coverage: UDP 8-Byte Header Structure, 16-Bit 1's Complement Checksum Calculation with Pseudo-Header, UDP Applications

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. UDP 8-Byte Header

• Source Port (16b), Destination Port (16b), Length (16b), Checksum (16b). Minimal overhead.

2. Checksum Calculation

• 16-bit 1's complement sum of header, data, and 12-byte IP Pseudo-Header. Result inverted at sender.

■ Worked Problem: UDP Checksum

• Given words: 0x66B4, 0x5555, 0x8F0C. Sum = 0x14B15 -> Wrap carry = 0x4B16 -> Invert = 0xB4E9 (Checksum).

■ KTU University High-Yield Exam Questions

• Q: Explain UDP 16-bit 1's complement checksum calculation with a worked example. (8 Marks)